Is bilinear value factorization a new SOTA on OGBench?

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Abstract

We swap the monolithic MLP value head of three OGBench reference agents (**GCIQL**, **HIQL**, **GCIVL**) for a rank-d bilinear factorization $Q(s,a,g) = \phi(s,a)^\top \psi(g)/\sqrt{d}$ and evaluate on **cube-single**, **antmaze-medium**, and **scene** (state obs). On cube, bilinear beats the published SOTA for *all three* methods once wired correctly (GCIQL **90.4** vs 68, GCIVL **79.2** vs 53, HIQL **41.3** vs 15); on antmaze the win is method-relative only, and on scene bilinear loses with every actor. The main failure mode is identified: the bilinear TD loss must not become the trainer of a representation the policy shares; HIQL recovers exactly when ψ is routed around its subgoal representation ζ.

1 The factorization, per algorithm

Offline GCRL setting per OGBench [1]: reward-free dataset $\mathcal{D}$, hindsight-relabelled goals $g \sim p(g \mid \tau)$ (current / future / random mixture, OGBench defaults), sparse reward $r(s,g) = \mathbf{1}[s \in S_g] - 1$, termination mask $m = \mathbf{1}[s \notin S_g]$. All value losses use the expectile loss [3] $\ell^2_\tau(x) = |\tau - \mathbf{1}[x<0]|\, x^2$, $\tau \in [0.5, 1)$; targets $\bar{V}, \bar{Q}$; 2-ensembles throughout.[1] We replace each value MLP with a rank-d (d=50) two-tower inner product, leaving losses, targets, and subgoal machinery untouched.

GCIVL (action-free goal-conditioned IQL [2]; AWR [4] actor):

$$\mathcal{L}_V^{\mathrm{IVL}} = \mathbb{E}\Big[\ell^2_\tau\big(r + \gamma m \, \min_i \bar{V}_i(s',g) - V(s,g)\big)\Big], \qquad \text{factorized:}\;\; V(s,g) = \tfrac{1}{\sqrt{d}}\,\phi(s)^\top\psi(g). \tag{1}$$

GCIQL (goal-conditioned IQL [3]; DDPG+BC [5] actor):

$$\mathcal{L}_V^{\mathrm{IQL}} = \mathbb{E}\Big[\ell^2_\tau\big(\min_i \bar{Q}_i(s,a,g) - V(s,g)\big)\Big], \qquad \mathcal{L}_Q = \mathbb{E}\Big[\textstyle\sum_i \big(Q_i(s,a,g) - r - \gamma m V(s',g)\big)^2\Big], \tag{2}$$

factorized in *both* heads: $Q_i(s,a,g) = \frac{1}{\sqrt{d}}\phi_i(s,a)^\top\psi_i(g)$, $V(s,g) = \frac{1}{\sqrt{d}}\phi_V(s)^\top\psi_V(g)$.

HIQL [2] trains one value with Eq. (1) but reads the goal through a learned, length-normalized subgoal representation $\zeta : \mathcal{S} \times \mathcal{G} \to \mathbb{R}^k$ (k=10, $\|\zeta\| = \sqrt{k}$):[2]

$$V(s,g) = f_\theta\big(s, \zeta(s,g)\big). \tag{3}$$

ζ is HIQL's "latent states as actions": the high-level policy is trained by AWR on $A^h = V(s_{t+n},g) - V(s_t,g)$ to output $z = \zeta(s_t, s_{t+n})$, and the low-level policy consumes z. On state observations the

[1] OGBench stabilizes the expectile regression with a double-DQN-style split (expectile weight from the target-net advantage, regressed difference from the online net); we keep it verbatim in every variant.

[2] The HIQL paper analyzes $V(s,\varphi(g))$ but adopts the state-conditioned $\varphi([g,s])$ (our $\zeta(s,g)$) as the empirically better variant; OGBench and we use exactly that.

low-level gradient into ζ is off by default, so **the value loss is ζ's sole trainer**. This makes the factorization ambiguous (ψ may eat g or $\zeta(s,g)$); we test three forms:

$$\text{(a) via shared rep (BVN-style [7]):} \quad V(s,g) = \tfrac{1}{\sqrt{d}}\,\phi(s)^\top \psi\big(\zeta(s,g)\big), \tag{4}$$

$$\text{(b) Schaul form [6]:} \quad V(s,g) = \tfrac{1}{\sqrt{d}}\,\phi(s)^\top \psi(g), \tag{5}$$

$$\text{(c) two-head split:} \quad V = \text{stock Eq. (3)}, \quad V^h(s,g) = \tfrac{1}{\sqrt{d}}\,\phi(s)^\top \psi(g)\big|_{\mathrm{sg}(\zeta)}. \tag{6}$$

In (a) the bilinear loss *replaces* the stock loss as ζ's trainer; in (b) ζ gets no value gradient (the low-level gradient is re-enabled to keep it trained); in (c) the stock V keeps TD, low level, and ζ-training, while a stop-gradient bilinear head serves only the high-level advantage. The GCIQL/GCIVL factorizations are Schaul-form by construction; HIQL's (a) is the exception, and it matters (§3).

Besides each method's stock actor we test a distilled *flow* actor (one-step flow-matching BC + Q-max distillation, à la FQL [8]); the actor turns out to be a first-order confound, so it gets separate rows.

2 What we evaluated

Table 1 is the evaluation record; protocol in the caption. Cube: bilinear beats the published SOTA for all three methods: GCIQL flow **90.4** (93.1 @1M; d=128/512: 94.2/94.6; a *domain* SOTA, best published across all eight OGBench methods is 68), GCIVL **67.5** (AWR) / **79.2** (flow), HIQL **32.8**–**41.3** via forms (b)/(c). Rank is not the knob: $d \in \{50, 128, 512\}$ is flat and d=5 cliffs to 12–15. Antmaze: **83.7** vs 71 is *method-relative* (published HIQL owns the domain at 96). Scene: bilinear loses with *both* actors: the one domain where the factorized value itself fails. Which actor unlocks bilinear flips by domain, so any claim must name (method, actor, domain):

GCIQL bilinear d=50	flow actor	DDPG+BC actor	stock MLP (best)
cube-single	90.4	51.3 $^\alpha$	71.7
antmaze-med	23.9	83.7	69.5
scene	29.2	25.4	51.6

3 Why: the shared-representation mechanism

The same factorization that buys GCIQL +22 and GCIVL +18–30 on cube moved HIQL by +1.8 and *hurt* its flow variant. The reason is in the algebra of §1: ζ is at once the value's goal encoder, the high-level action space, and the low-level goal input, and form (4) silently swaps its only training signal from the stock TD loss to the bilinear one. GCIQL/GCIVL have no such shared module. A linear probe of ζ @500k confirms: no rank collapse (participation ratio rises), but global geometry degrades under (4) (goal-xyz R^2 on random pairs $0.14 \to 0.08$, rep-vs-goal-distance Spearman $0.115 \to 0.06$) while subgoal-scale structure is spared. Cutting the value$\to\zeta$ path, by (5) or (6), is the fix (Table 2): every cell where the bilinear loss trains ζ is flat-to-worse and capacity ($k\,10 \to 50$) does not rescue it; every cell where it does not is a 2–2.7× win over the published 15, and the flow cell flips from bilinear's worst (14.5) to its best (41.3°). The matched-protocol antmaze control (MLP 95.3 vs form-(a) 95.9, 8 seeds @1M) shows the warp costs nothing at that domain's ceiling, which is why it was invisible where HIQL shines. The design rule: **the bilinear head must not train a representation the policy shares**; when in doubt, give it a stop-gradient path. Scene, where bilinear fails for GCIQL with both actors, plausibly needs the value to compose multiple object-to-object interactions, the one setting where a rank-50 product of a state tower and a goal tower is the wrong bias; the (5)-vs-(4) axis is untested there and is the obvious next probe.

Method	Value head	Actor	cube-single	antmaze-med	scene
GCIQL	OGBench (pub.)	native	68 ± 6	71 ± 4	51 ± 4
	MLP (native)	DDPG+BC	70.3 ± 4.7	69.5 ± 4.4	51.6 ± 3.6
	MLP (native)	flow	71.7 ± 4.9	33.3 ± 5.2	*not run*
	bilinear d=50	flow	90.4 ± 3.4	23.9 ± 1.6	29.2 ± 3.2
	bilinear d=50	DDPG+BC	51.3 ± 8.0 $^{\alpha}$	83.7 ± 0.7	25.4 ± 0.3 ^{s}
HIQL	OGBench (pub.)	native	15 ± 3	96 ± 1	38 ± 3
	MLP (native)	hier. AWR	16.7 ± 5.8	95.3 ± 0.6 ^{m}	*not run*
	bilinear Eq. (4)	hier. AWR	18.5 ± 0.2	95.9 ± 0.5 ^{m}	*not run*
	bilinear Eq. (5)	hier. AWR	32.8 ± 0.4	*not run*	*not run*
	bilinear Eq. (6)	hier. AWR	34.3 ± 5.4 $^{\circ}$	*not run*	*not run*
	bilinear Eq. (6)	flow	41.3 ± 2.3 $^{\circ}$	*not run*	*not run*
GCIVL	OGBench (pub.)	native	53 ± 4	72 ± 8	42 ± 4
	MLP (native)	AWR	49.6 ± 2.0	*not run*	*not run*
	MLP (native)	flow	59.1 ± 4.2	*not run*	*not run*
	bilinear d=50	AWR	67.5 ± 8.7	*not run*	*not run*
	bilinear d=50	flow	79.2 ± 0.4 †	*not run*	*not run*

Table 1: **Does bilinear factorization beat SOTA?** State obs; `evaluation/overall_success` (%) over 5 eval goals, mean $\pm$ SD, n=3 seeds @500k unless noted. Green=local, orange=OGBench paper (8 seeds @1M); **bold green** = beats the published number for that method/domain. ^{m}matched protocol: 8 seeds, 1M, avg last-3 evals. $^{\circ}$read @400k, runs finishing. $^{\alpha}\alpha$=0.3 vs stock α=1.0 (confounded, rerun owed). ^{s}n=2 (third seed @300k: 20.8). †healthy batch; an earlier batch died at init and had contaminated this cell as "47.5 ± 38.8". HIQL cube cells use the tuned `subgoal_steps`=10. *not run* = not run.

Caveats.

- Local = 500k/3 seeds vs published 1M/8 seeds; where both exist 500k *under*states bilinear ($90.4 \rightarrow 93.1$), but the winning cells still owe a matched 8-seed 1M confirmation.
- $^{\circ}$ cells are @400k with runs finishing; the ordering is already fixed by the fully-final Schaul cell (32.8 @500k).
- Confounds: bilinear d=50 has 2.06$\times$ the value-head params of the MLP (control owed); cube DDPG+BC ran α=0.3 ($^{\alpha}$); scene is untested on the (5)-vs-(4) axis.
- Not run: GCIVL beyond cube, HIQL mechanism cells beyond cube, pixels.

Provenance. wandb `amsks/factored-fb` (pulled 2026-07-06, refreshed from run-local `eval.csv` 2026-07-07), metric `evaluation/overall_success`; published numbers from OGBench [1], Table 2.

References

[1] S. Park, K. Frans, B. Eysenbach, and S. Levine. OGBench: Benchmarking offline goal-conditioned RL. In *ICLR*, 2025. arXiv:2410.20092.

[2] S. Park, D. Ghosh, B. Eysenbach, and S. Levine. HIQL: Offline goal-conditioned RL with latent states as actions. In *NeurIPS*, 2023. arXiv:2307.11949.

[3] I. Kostrikov, A. Nair, and S. Levine. Offline reinforcement learning with implicit Q-learning. In *ICLR*, 2022. arXiv:2110.06169.

[3] Earlier drafts ran `subgoal_steps`=25 (MLP 24.6, bilinear 15.9) and concluded "bilinear doesn't help HIQL"; at the tuned sg=10 the baseline reproduces the published 15 and that conclusion is retired.

Cell	who trains ζ	high-level advantage from	success
MLP (baseline), Eq. (3)[3]	MLP value	MLP V	16.7 ± 5.8
bilinear via shared rep, Eq. (4)	**bilinear value**	bilinear V	18.5 ± 0.2
$+\ \zeta$ dim k: $10 \to 50$	**bilinear value**	bilinear V	$24.3 \pm 1.0^\circ$
bilinear, Schaul $\psi(g)$, Eq. (5)	low actor	bilinear V	32.8 ± 0.4
two-head split, Eq. (6)	MLP value	stop-grad bilinear head	$34.3 \pm 5.4^\circ$
flow actors: MLP	MLP value	MLP V	22.7 ± 5.2
flow, bilinear via shared rep	**bilinear value**	bilinear V	14.5 ± 2.6
flow, two-head split	MLP value	stop-grad bilinear head	$41.3 \pm 2.3^\circ$

Table 2: **HIQL mechanism ladder** (cube-single, `subgoal_steps`=10, d=50, 3 seeds @500k; $^\circ$=@400k, finishing). **Bold "who trains ζ"** = the bilinear TD loss shapes the shared subgoal representation.

[4] X. B. Peng, A. Kumar, G. Zhang, and S. Levine. Advantage-weighted regression: Simple and scalable off-policy reinforcement learning. arXiv:1910.00177, 2019.

[5] S. Fujimoto and S. S. Gu. A minimalist approach to offline reinforcement learning. In *NeurIPS*, 2021. arXiv:2106.06860.

[6] T. Schaul, D. Horgan, K. Gregor, and D. Silver. Universal value function approximators. In *ICML*, 2015.

[7] Z.-W. Hong, G. Yang, and P. Agrawal. Bilinear value networks. In *ICLR*, 2023. arXiv:2204.13695.

[8] S. Park, Q. Li, and S. Levine. Flow Q-learning. In *ICML*, 2025. arXiv:2502.02538.